Youjing Yu

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Education

Computational and Biological Learning Lab, University of Cambridge *Oct. 2023 - PRESENT*
ENGINEERING PhD in Training
Supervisor: Professor Guillaume Hennequin
Advisor: Dr Flavia Mancini

Trinity College, University of Cambridge *Oct. 2019 – Jun.2023*
ENGINEERING Bachelor of Arts, Master of Engineering
First Year Grades: *Unofficial due to COVID-19*; 85% Overall.
Second Year Grades: *Unofficial due to COVID-19*; 77% Overall.
Third Year Grades: 78% Overall;
ranked 19/285 (top 6%) of general engineering;.
Fourth Year (Masters) Grades: 80% Overall;
ranked 3/285 (top 1%) of general engineering;
ranked top of Information and Computer engineering.

Hwa Chong Institution, Singapore *Jan. 2017 - Dec. 2018*
Singapore A level Physics (A), Mathematics (A), Chemistry (A), Economics (A), Modern Physics (Distinction)

Awards & Prizes

Cambridge University Research Scholar - Oct 2023
Trinity College Internal Graduate Studentship - Aug 2023;
Trinity College Mathison Prize - Aug 2023;
Trinity College Richard and Valery Garrett Prize - Aug 2023;
Trinity College Prince of Wales Student Award - July 2023;
Trinity College Examination Prize - July 2023;
Department of Engineering AT&T Prize - June 2023;
Trinity College Senior Scholar - July 2022;
Trinity College Examination Prize - July 2022;
Trinity College Summer Studentship - Mar 2022;
Trinity College Senior Scholar - July 2021;
Trinity College Examination Prize - July 2021;
Cambridge Trust Overseas Scholarship - Mar 2019;
Trinity College Overseas Bursary - Mar 2019;
Hwa Chong Diploma with Distinction - 2018;
Singapore Science and Engineering Fair Silver Award - March 2018;

Publications

- **Yu, Y.**, Shi, Z., & Lipani, A. (2024, June). Understanding Users' Confidence in Spoken Queries for Conversational Search Systems. In *International Conference on Engineering Applications of Neural Networks* (pp. 405-418). Cham: Springer Nature Switzerland.
- **Yu, Y.**, Xia, R., Ma, Q., Lengyel, M., & Hennequin, G. (2024). Second-order forward-mode optimization of recurrent neural networks for neuroscience. *Advances in Neural Information Processing Systems*, 37, 92101-92129.
- Artemev, A., Xia, R., Boyd, B. M., **Yu, Y.**, Dangel, F., Hennequin, G., & Bernacchia, A. (2026). Exploiting weight-space symmetries for approximating curvature. In *Proceedings of the 43rd International Conference on Machine Learning, Proceedings of Machine Learning Research, 306, 3856–3890.*
- Walker, S., **Yu, Y.**, Atkins, N. *A Data-Driven Approach for Generalisable Prediction of Hot Gas Ingestion Across Unseen Geometries*. (under review)

Experience

Machine Learning Researcher Intern **MediaTek**
Machine Learning Team *Oct.2025-Feb.2026*

- Contributed to research on exploiting weight-space symmetries to construct efficient, structured Hessian approximations from training gradients.

Machine Learning Engineer Intern **Myrtle.ai**
Machine Learning Research Team *Jul.2025-Sep.2025*

- Developed a low-latency real-time Automatic Speech Translation (AST) system deployable on CPU, GPU and FPGA.

Masters Student **University of Cambridge**
Department of Engineering *Oct.2022-Jun.2023*

- Developed a webcam-based eye-tracking system using TensorFlow FaceMesh to estimate on-screen gaze.
- Extended WebGazer by applying Gaussian Process regression with a novel kernel to map eye images to eyeball rotation angles. 22% reduction in gaze-prediction error compared to the original WebGazer.

Research Intern at the Department of Computer Science **University College London**
Department of Computer Science *Dec.2021-Oct.2022*

- Developed UNSURE, a Spotify Podcast dataset of user queries and human confidence labels.
- Developed a transformer-based multimodal network combining text and audio to estimate questioners' topic-specific knowledge confidence.

Research Intern **University of Cambridge**
Department of Engineering *Jul.2021-Oct 2021*

- Developed a Flask-based NLP corpus analysis tool featuring two bias-detection algorithms to visualise and explore gender bias interactively.
- Implemented a novel debiasing feature and natural-language query processing.

Research Intern **University of Cambridge**
MAGIC Research Group *Mar.2021- Apr.2021*

- Developed neural networks to identify car plate numbers from videos with unsupervised training.

Student Researcher at **University of Cambridge**
Spacept Hackbridge *Oct.2020-Dec 2020*

- Built a computer vision change detector system for identifying and predicting building damage and forest fires from satellite images.

Student Assistant Researcher **University of Hong Kong**
Department of Electrical and Electronic Engineering *Jul.2020-Sep.2020*

- Developed deep learning models in TensorFlow for holography image generation and image-to-image translation, implementing and evaluating CNN, ResNet, Wide ResNet, DenseNet, and SqueezeNet architectures.

Student Researcher **National University of Singapore**
Department of Building *Mar.2017-Jan.2018*

- Prepared nano encapsulated organic particles in the wet lab.
- Our Cadmium Orthostannate nanoparticles with 3-(mercaptopropyl)trimethoxysilane suspended in organic solution were shown to have effective thermal insulation abilities.

Talks & Posters

- Information Engineering Conference at University of Cambridge: "*Learning without forgetting: how plasticity and dynamics enable continual learning*." Oct. 2026.
- Poster at Computational and Systems Neuroscience (COSYNE). Mar. 2026.
- Poster at Statistical Analysis of Neural Data (SAND). June. 2025.

Teaching

Trinity College — **University of Cambridge**
Oct.2023 – May.2025

- Engineering Tripos 1A Mathematics

Extra-curricular Activities

Journal Club Coordinator — **University of Cambridge**
Biological Learning Group — *Oct.2024 – Sep.2026*

- Organising student-led lab meetings and journal clubs.

President — **University of Cambridge**
Trinity College Badminton Club — *Oct.2022 – Jun.2023*

- Overseeing Women's Team, Men's First, Second and Third Team.
- Organise coaching, league matches, cuppers matches, field court bookings, stash designs and orders.

Member — **Singapore**
Hwa Chong Girls' Tennis Team — *Jan.2017 – Dec.2018*

- Represented Hwa Chong Junior College in inter-collegiate tennis matches.
- **First Runner-Up** in 2017; Second Runner-Up in 2018.

Member — **Singapore**
Hwa Chong Junior College Chinese Orchestra — *Jan.2017 – June.2017*

- Represented Hwa Chong Junior College Chinese Orchestra.
- **Gold Award** at Singapore Youth Festival.

Member — **Singapore**
Son-rise Singapore — *Jan.2017 – Dec.2018*

- Responsible for interacting and playing with an autistic child four hours a week. Taught the child basic words and maths.
- This experience taught me the value of empathy and the importance of always respecting and helping people living on the margins of society, which I am highly grateful for. I have also developed a positive mindset and a passion for serving.

Member — **Singapore**
Anderson Secondary School Choir Member — *Jan.2015 – Dec.2016*

- Represented Anderson Secondary School Choir
- **Gold Award** at Singapore Youth Festival.

Referees

Dr Flavia Mancini
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Department of Engineering, Cambridge

Dr Guillaume Hennequin
gjeh2@cam.ac.uk
Department of Engineering, Cambridge